

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250282032

Kind Code

A1

Publication Date

September 11, 2025

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Wrench Socket

Abstract

A wrench socket, comprising a body having an axis, wherein the body is provided with a small-diameter section and a large-diameter section connected to the small-diameter section along the axis. The large-diameter section of the body has a side surface. A plurality of groove portions are arranged at intervals around the side surface of the large-diameter section. Each groove portion is formed on one end or both ends of the side surface along the axis. Each groove portion is tapered along the axis and has an included angle. Thereby, the wrench socket provides both anti-slip and anti-rolling capabilities, avoiding operational inconvenience and shortening operation time.

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Family ID: 96261940

Appl. No.: 19/074575

Filed: March 10, 2025

Foreign Application Priority Data

TW 113108619

Mar. 08, 2024

Publication Classification

Int. Cl.: B25B13/06 (20060101)

U.S. Cl.:

CPC B25B13/065 (20130101);

Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates to a socket, particularly to a wrench socket.

DESCRIPTION OF THE RELATED ART

[0002] Conventionally, wrench sockets have a smooth exterior surface (see, for example, FIG. 1 of Taiwan Patent Publication No. TWM558688U, which discloses an exemplary embodiment). When operating a wrench socket, the surface must be gripped with the palm of the hand. However, natural sweat from the palm itself or oil stains adhered to the palm make it difficult to grip the smooth surface. Moreover, when the wrench socket is placed at a workplace (such as a machine), the smooth surface is prone to rolling. These issues cause the wrench socket to drop easily, resulting in operational inconvenience and prolonged operation time.

SUMMARY OF THE INVENTION

[0003] An object of the present invention is to provide a wrench socket to solve the problems of the prior art as mentioned above.

[0004] According to one embodiment of the present invention, a wrench socket is provided, comprising: a body having an axis and a side surface, wherein a plurality of groove portions are arranged at intervals around the side surface, each of the plurality of groove portions is formed on the side surface between two ends of the body along the axis, and each of the plurality of groove portions is tapered along the axis and has an included angle.

[0005] The present invention, by virtue of the groove portions, provides the wrench socket with both anti-slip and anti-rolling capabilities, thereby avoiding operational inconvenience and shortening operation time.

[0006] Numerous objects, features and advantages of the invention will be readily apparent upon a reading of the following detailed description of embodiments of the invention when taken in conjunction with the accompanying drawings. However, the drawings employed herein are for the purpose of descriptions and should not be regarded as limiting.

Description

BRIEF DESCRIPTION OF DRAWINGS

[0007] The above objects and advantages of the invention will become more readily apparent to those ordinarily skilled in the art after reviewing the following detailed description and accompanying drawings, in which:

[0008] FIG. 1 is a perspective view of a first embodiment of the wrench socket of the present invention.

[0009] FIG. 2A is a first cross-sectional view of the groove portion of the wrench socket of the present invention along the X direction.

[0010] FIG. 2B is a second cross-sectional view of the groove portion of the wrench socket of the present invention along the X direction.

[0011] FIG. 2C is a third cross-sectional view of the groove portion of the wrench socket of the present invention along the X direction.

[0012] FIG. 3A is a first cross-sectional view of the groove portion of the wrench socket of the present invention along the Y direction.

[0013] FIG. 3B is a second cross-sectional view of the groove portion of the wrench socket of the present invention along the Y direction.

[0014] FIG. 3C is a third cross-sectional view of the groove portion of the wrench socket of the present invention along the Y direction.

[0015] FIG. 4 is a schematic view showing the combination of the wrench socket of the present invention with a driving member and a driven member.

[0016] FIG. 5 is a schematic view showing the anti-rolling feature of the wrench socket of the present invention.

[0017] FIG. 6 is a schematic view showing the anti-slip feature of the wrench socket of the present invention.

[0018] FIG. 7 is a view showing a first operational state of the wrench socket of the present invention.

[0019] FIG. 8 is a view showing a second operational state of the wrench socket of the present invention.

[0020] FIG. 9 is a view showing a first variant of the wrench socket of the present invention shown in FIG. 1.

[0021] FIG. 10 is a perspective view of a second embodiment of the wrench socket of the present invention.

[0022] FIG. 11 is a front view and a rear view of the wrench socket shown in FIG. 10.

[0023] FIG. 12 is a left side view and a right side view of the wrench socket shown in FIG. 10.

[0024] FIG. 13 is a top view of the wrench socket shown in FIG. 10.

[0025] FIG. 14 is a bottom view of the wrench socket shown in FIG. 10.

[0026] FIG. 15 is a view showing a first variant of the wrench socket of the present invention shown in FIG. 10.

[0027] FIG. 16 is a view showing a second variant of the wrench socket of the present invention shown in FIG. 10.

[0028] FIG. 17 is a view showing a third variant of the wrench socket of the present invention shown in FIG. 10.

[0029] FIG. 18 is a perspective view of a third embodiment of the wrench socket of the present invention.

[0030] FIG. 19 is a front view and a rear view of the wrench socket shown in FIG. 18.

[0031] FIG. 20 is a left side view and a right side view of the wrench socket shown in FIG. 18.

[0032] FIG. 21 is a top view of the wrench socket shown in FIG. 18.

[0033] FIG. 22 is a bottom view of the wrench socket shown in FIG. 18.

[0034] FIG. 23 is a view showing a first variant of the wrench socket of the present invention shown in FIG. 18.

[0035] FIG. 24 is a view showing a second variant of the wrench socket of the present invention shown in FIG. 18.

[0036] FIG. 25 is a view showing a third variant of the wrench socket of the present invention shown in FIG. 18.

[0037] FIG. 26 is another perspective view of the wrench socket shown in FIG. 18.

[0038] FIG. 27 is a front view of the wrench socket shown in FIG. 26.

[0039] FIG. 28 is a rear view of the wrench socket shown in FIG. 26.

[0040] FIG. 29 is a left side view of the wrench socket shown in FIG. 26.

[0041] FIG. 30 is a right side view of the wrench socket shown in FIG. 26.

[0042] FIG. 31 is a top view of the wrench socket shown in FIG. 26.

[0043] FIG. 32 is a bottom view of the wrench socket shown in FIG. 26.

[0044] FIG. 33 is a view showing a fourth variant of the wrench socket of the present invention shown in FIG. 18.

[0045] FIG. 34 is a front view of the wrench socket shown in FIG. 33.

[0046] FIG. 35 is a rear view of the wrench socket shown in FIG. 33.

[0047] FIG. 36 is a left side view of the wrench socket shown in FIG. 33.

[0048] FIG. 37 is a right side view of the wrench socket shown in FIG. 33.

[0049] FIG. 38 is a top view of the wrench socket shown in FIG. 33.

[0050] FIG. 39 is a bottom view of the wrench socket shown in FIG. 33.

[0051] FIG. 40 is another perspective view of the wrench socket shown in FIG. 1.

[0052] FIG. **41** is a front view of the wrench socket shown in FIG. **1**.
[0053] FIG. **42** is a rear view of the wrench socket shown in FIG. **1**.
[0054] FIG. **43** is a left side view of the wrench socket shown in FIG. **1**.
[0055] FIG. **44** is a right side view of the wrench socket shown in FIG. **1**.
[0056] FIG. **45** is a top view of the wrench socket shown in FIG. **1**.
[0057] FIG. **46** is a bottom view of the wrench socket shown in FIG. **1**.
[0058] FIG. **47** is a perspective view of a fourth embodiment of the wrench socket of the present invention.
[0059] FIG. **48** is a front view of the wrench socket shown in FIG. **47**.
[0060] FIG. **49** is a rear view of the wrench socket shown in FIG. **47**.
[0061] FIG. **50** is a left side view of the wrench socket shown in FIG. **47**.
[0062] FIG. **51** is a right side view of the wrench socket shown in FIG. **47**.
[0063] FIG. **52** is a top view of the wrench socket shown in FIG. **47**.
[0064] FIG. **53** is a bottom view of the wrench socket shown in FIG. **47**.
[0065] FIG. **54** is a view showing a first variant of the wrench socket of the present invention shown in FIG. **47**.
[0066] FIG. **55** is a front view of the wrench socket shown in FIG. **54**.
[0067] FIG. **56** is a rear view of the wrench socket shown in FIG. **54**.
[0068] FIG. **57** is a left side view of the wrench socket shown in FIG. **54**.
[0069] FIG. **58** is a right side view of the wrench socket shown in FIG. **54**.
[0070] FIG. **59** is a top view of the wrench socket shown in FIG. **54**.
[0071] FIG. **60** is a bottom view of the wrench socket shown in FIG. **54**.
[0072] FIG. **61** is a view showing a fifth variant of the wrench socket of the present invention shown in FIG. **18**.
[0073] FIG. **62** is a front view of the wrench socket shown in FIG. **61**.
[0074] FIG. **63** is a rear view of the wrench socket shown in FIG. **61**.
[0075] FIG. **64** is a left side view of the wrench socket shown in FIG. **61**.
[0076] FIG. **65** is a right side view of the wrench socket shown in FIG. **61**.
[0077] FIG. **66** is a top view of the wrench socket shown in FIG. **61**.
[0078] FIG. **67** is a bottom view of the wrench socket shown in FIG. **61**.
[0079] FIG. **68** is a view showing a sixth variant of the wrench socket of the present invention shown in FIG. **18**.
[0080] FIG. **69** is a front view of the wrench socket shown in FIG. **68**.
[0081] FIG. **70** is a rear view of the wrench socket shown in FIG. **68**.
[0082] FIG. **71** is a left side view of the wrench socket shown in FIG. **68**.
[0083] FIG. **72** is a right side view of the wrench socket shown in FIG. **68**.
[0084] FIG. **73** is a top view of the wrench socket shown in FIG. **68**.
[0085] FIG. **74** is a bottom view of the wrench socket shown in FIG. **68**.
[0086] FIG. **75** is a view showing a second variant of the wrench socket of the present invention shown in FIG. **1**.
[0087] FIG. **76** is a front view of the wrench socket shown in FIG. **75**.
[0088] FIG. **77** is a rear view of the wrench socket shown in FIG. **75**.
[0089] FIG. **78** is a left side view of the wrench socket shown in FIG. **75**.
[0090] FIG. **79** is a right side view of the wrench socket shown in FIG. **75**.
[0091] FIG. **80** is a top view of the wrench socket shown in FIG. **75**.
[0092] FIG. **81** is a bottom view of the wrench socket shown in FIG. **75**.
[0093] FIG. **82** is a view showing a second variant of the wrench socket of the present invention shown in FIG. **47**.
[0094] FIG. **83** is a front view of the wrench socket shown in FIG. **82**.
[0095] FIG. **84** is a rear view of the wrench socket shown in FIG. **82**.

[0096] FIG. **85** is a left side view of the wrench socket shown in FIG. **82**.
[0097] FIG. **86** is a right side view of the wrench socket shown in FIG. **82**.
[0098] FIG. **87** is a top view of the wrench socket shown in FIG. **82**.
[0099] FIG. **88** is a bottom view of the wrench socket shown in FIG. **82**.
[0100] FIG. **89** is a view showing a third variant of the wrench socket of the present invention shown in FIG. **47**.
[0101] FIG. **90** is a front view of the wrench socket shown in FIG. **89**.
[0102] FIG. **91** is a rear view of the wrench socket shown in FIG. **89**.
[0103] FIG. **92** is a left side view of the wrench socket shown in FIG. **89**.
[0104] FIG. **93** is a right side view of the wrench socket shown in FIG. **89**.
[0105] FIG. **94** is a top view of the wrench socket shown in FIG. **89**.
[0106] FIG. **95** is a bottom view of the wrench socket shown in FIG. **89**.
[0107] FIG. **96** is a view showing a seventh variant of the wrench socket of the present invention shown in FIG. **18**.
[0108] FIG. **97** is a front view of the wrench socket shown in FIG. **96**.
[0109] FIG. **98** is a rear view of the wrench socket shown in FIG. **96**.
[0110] FIG. **99** is a left side view of the wrench socket shown in FIG. **96**.
[0111] FIG. **100** is a right side view of the wrench socket shown in FIG. **96**.
[0112] FIG. **101** is a top view of the wrench socket shown in FIG. **96**.
[0113] FIG. **102** is a bottom view of the wrench socket shown in FIG. **96**.
[0114] FIG. **103** is a view showing an eighth variant of the wrench socket of the present invention shown in FIG. **18**.
[0115] FIG. **104** is a front view of the wrench socket shown in FIG. **103**.
[0116] FIG. **105** is a rear view of the wrench socket shown in FIG. **103**.
[0117] FIG. **106** is a left side view of the wrench socket shown in FIG. **103**.
[0118] FIG. **107** is a right side view of the wrench socket shown in FIG. **103**.
[0119] FIG. **108** is a top view of the wrench socket shown in FIG. **103**.
[0120] FIG. **109** is a bottom view of the wrench socket shown in FIG. **103**.
[0121] FIG. **110** is a view showing a third variant of the wrench socket of the present invention shown in FIG. **1**.
[0122] FIG. **111** is a front view of the wrench socket shown in FIG. **110**.
[0123] FIG. **112** is a rear view of the wrench socket shown in FIG. **110**.
[0124] FIG. **113** is a left side view of the wrench socket shown in FIG. **110**.
[0125] FIG. **114** is a right side view of the wrench socket shown in FIG. **110**.
[0126] FIG. **115** is a top view of the wrench socket shown in FIG. **110**.
[0127] FIG. **116** is a bottom view of the wrench socket shown in FIG. **110**.
[0128] FIG. **117** is a view showing a fourth variant of the wrench socket of the present invention shown in FIG. **47**.
[0129] FIG. **118** is a front view of the wrench socket shown in FIG. **117**.
[0130] FIG. **119** is a rear view of the wrench socket shown in FIG. **117**.
[0131] FIG. **120** is a left side view of the wrench socket shown in FIG. **117**.
[0132] FIG. **121** is a right side view of the wrench socket shown in FIG. **117**.
[0133] FIG. **122** is a top view of the wrench socket shown in FIG. **117**.
[0134] FIG. **123** is a bottom view of the wrench socket shown in FIG. **117**.
[0135] FIG. **124** is a view showing a fifth variant of the wrench socket of the present invention shown in FIG. **47**.
[0136] FIG. **125** is a front view of the wrench socket shown in FIG. **124**.
[0137] FIG. **126** is a rear view of the wrench socket shown in FIG. **124**.
[0138] FIG. **127** is a left side view of the wrench socket shown in FIG. **124**.
[0139] FIG. **128** is a right side view of the wrench socket shown in FIG. **124**.

[0140] FIG. **129** is a top view of the wrench socket shown in FIG. **124**.

[0141] FIG. **130** is a bottom view of the wrench socket shown in FIG. **124**.

DETAILED DESCRIPTION OF THE INVENTION

[0142] Reference will now be made in detail to the preferred embodiments of the present invention, examples of which are illustrated in the accompanying drawings. While the invention will be described in conjunction with the preferred embodiments, it will be understood that they are not intended to limit the invention to these embodiments. On the contrary, the invention is intended to cover alternatives, modifications and equivalents, which may be included within the spirit and scope of the invention as defined by the appended claims.

[0143] Referring to FIGS. **1** to **130**, the present invention provides a wrench socket including a body **10** in a first embodiment (refer to FIGS. **1** to **9**), which will be described as follows. The body **10** is formed by forging, but is not limited to forging. The body **10** has an approximately cylindrical appearance.

[0144] Referring to FIG. **1**, the body **10** has an axis **10A**. The body **10** is provided axially along the axis **10A** with a small-diameter section **11** and a large-diameter section **12** connected to the small-diameter section **11**. The maximum outer diameter of the small-diameter section **11** is smaller than the maximum outer diameter of the large-diameter section **12**. Furthermore, a connecting section **13** is provided between the small-diameter section **11** and the large-diameter section **12** to connect the sections having different outer diameters. The connecting section **13** has a connecting angle, which may be selected from curved corner, chamfered corner, and right angle, or combinations thereof. The inclined angle includes acute angle (greater than 0° and less than 90° and obtuse angle (greater than 90° and less than 180°).

[0145] Referring to FIG. **1**, the small-diameter section **11** is provided with a force-receiving part **111** from one end face of the body **10** along the axis **10A**, and has a force-receiving opening **112** recessed toward the large-diameter section **12**. The large-diameter section **12** is provided with a force-applying part **121** from another end face of the body **10** along the axis **10A**, and has a force-applying opening **122** recessed toward the small-diameter section **11**.

[0146] The force-receiving part **111** and the force-applying part **121** may be non-communicating slots or may be communicating holes, but are not limited to slots or holes. FIG. **5** of Taiwan Patent Publication No. TWM558688U shows an example of slots. FIG. **9** of Taiwan Patent Publication No. TWM558688U and FIG. **1** of Taiwan Patent Publication No. TW249372U show examples of holes.

[0147] Referring to FIGS. **1** and **4**, the force-receiving part **111** combines with a driving member E through the force-receiving opening **112**. One example of the driving member E may be a wrench, but is not limited to a wrench. The force-applying part **121** combines with a driven member R through the force-applying opening **122**. One example of the driven member R may be a screw, but is not limited to a screw. When the user applies force to the driving member E, the force-receiving part **111** of the small-diameter section **11** rotates under force, causing the force-applying part **121** of the large-diameter section **12** to apply force to the driven member R, thereby causing the driven member R to rotate under force.

[0148] Referring to FIG. **1**, the outer wall of the large-diameter section **12** of the body **10** has a side surface **12A** surrounding the axis **10A**. A plurality of groove portions **20** are arranged at intervals around the side surface **12A** of the large-diameter section **12**. Each groove portion **20** is formed on one end of the side surface **12A** along the axis **10A**. The groove portion **20** is approximately bullet-shaped.

[0149] Referring to FIG. **1**, the side surface **12A** further defines a first end **12A1** and a second end **12A2**. Each groove portion **20** starts from the second end **12A2** (that is to say, each groove portion **20** contacts the second end **12A2**) and terminates before the first end **12A1** (that is to say, each groove portion **20** does not contact the first end **12A1**). A dividing line **12A3** is further defined between the first end **12A1** and the second end **12A2**. The dividing line **12A3** may be a midline,

with the midline being equidistant from the first end **12A1** and the second end **12A2**. Each groove portion **20** further terminates at the dividing line **12A3** (that is to say, each groove portion **20** contacts the dividing line **12A3**) or terminates before the dividing line **12A3** (not shown, that is to say, each groove portion **20** does not contact the dividing line **12A3**).

[0150] Each groove portion **20** is gradually tapered along the axis **10A** and has an included angle **20A** (refer to FIG. 1). Each groove portion **20** is gradually tapered along the axis **10A** from the front end toward the rear end, and has at the rear end an included angle **20A** that becomes narrower in width (refer to FIG. 1, viewed axially from the front end toward the rear end) and shallower in depth (refer to FIG. 2, viewed axially from the front end toward the rear end). When viewed in cross-section along the X direction, the included angle **20A** may be selected from rounded corner (refer to FIGS. 2A and 2B), curved corner (refer to FIG. 2C), and chamfered corner, or combinations thereof. The chamfered corner includes acute angle and obtuse angle to adapt to different usage requirements. The groove portion **20** provides the body **10** with anti-rolling capability (refer to FIG. 5) and anti-slip capability (refer to FIG. 6). Different angles of the included angle **20A** of the groove portion **20** produce different effects. For example, acute angle can enhance anti-slip effect in specific directions; right angle provides definite anti-rolling stop points; obtuse angle provides larger contact area; rounded corner can reduce stress concentration; curved corner provides progressive anti-slip effect. The present invention may also employ combinations of the above angle types. For example, the groove portion **20** may employ different angle designs at different positions, thereby providing anti-slip and anti-rolling effects under different working conditions.

[0151] Referring to FIGS. 1 and 2, each groove portion **20** further forms, along the axis **10A** from the front end toward the rear end, a straight section **21** and a tapered section **22** connecting to the straight section **21**. The tapered section **22** has an included angle **20A**. When viewed in cross-section along the Y direction, the straight section **21** may be selected from arc-shaped (refer to FIG. 3A), U-shaped (refer to FIG. 3B), and A-shaped (refer to FIG. 3C), or combinations thereof. The Y direction and the X direction intersect each other, and further, the Y direction and the X direction are orthogonal or perpendicular to each other.

[0152] Each groove portion **20** of the large-diameter section **12** may also be formed on another end of the side surface **12A** along the axis **10A**. Each groove portion **20** may also start from the first end **12A1** (not shown, that is to say, each groove portion **20** contacts the first end **12A1**) and terminate before the second end **12A2** (not shown, that is to say, each groove portion **20** does not contact the second end **12A2**). Each groove portion **20** further terminates at the dividing line **12A3** (not shown, that is to say, each groove portion **20** contacts the dividing line **12A3**) or terminates before the dividing line **12A3** (not shown, that is to say, each groove portion **20** does not contact the dividing line **12A3**).

[0153] Each groove portion **20** of the large-diameter section **12** may also be simultaneously formed at both ends of the side surface **12A** along the axis **10A**. The starting and terminating positions of each groove portion **20** are as described above. It should be noted that the groove portions **20** at the two ends of the side surface **12A** do not contact each other.

[0154] The small-diameter section **11** may also have another force-receiving part **113** provided on a side surface of the small-diameter section **11** surrounding the axis **10A** (refer to FIG. 1). The force-receiving part **113** combines with another driving member that has a driving opening (refer to the example shown in FIG. 5 of Taiwan Patent Publication No. TWM243300U). The driving member may be a wrench. The force-receiving parts **111** and **113** may be simultaneously provided on the small-diameter section **11** (refer to FIG. 1), or alternatively, only one of the force-receiving parts **111** and **113** may be provided on the small-diameter section **11** (not shown, that is to say, selecting either force-receiving part **111** or **113**).

[0155] FIGS. 7 to 9 show different operational states of the body **10** and different variant configurations of the body **10** of the first embodiment. The first embodiment may also refer to

FIGS. 40 to 46.

[0156] Referring to **FIGS. 10 to 17**, the second embodiment is described as follows. Refer to **FIG. 10**: the differences from the first embodiment lie in the variations of the small-diameter section **11'**, large-diameter section **12'**, and connecting section **13'** of the body **10'**. The groove portion **20** maintains an approximately bullet-shaped configuration. The body **10'** in perspective view (refer to **FIG. 10**), the body **10'** in front view (refer to **FIG. 11**, with the rear view being identical to the front view, thus omitted), **FIG. 11** uses break lines to indicate that the small-diameter section **11'** and large-diameter section **12'** have consistent shape, structure, or regular variation along their length. In left side view (refer to **FIG. 12**, with the right side view being identical to the left side view, thus omitted), in top view (refer to **FIG. 13**), in bottom view (refer to **FIG. 14**). Additionally, different variant configurations of the body **10'** incorporate the approximately bullet-shaped groove portion **20** (refer to **FIGS. 15, 16, 17**).

[0157] Referring to **FIGS. 18 to 25**, the third embodiment is described as follows. Refer to **FIG. 18**: the difference from the second embodiment lies in the variation of the groove portion **20'** to an approximately capsule-shaped configuration, while the body **10'** remains unchanged. The body **10'** in perspective view (refer to **FIG. 18**), in front view (refer to **FIG. 19**, with the rear view being identical to the front view, thus omitted), **FIG. 19** uses break lines to indicate that the small-diameter section **11'** and large-diameter section **12'** have consistent shape, structure, or regular variation along their length. In left side view (refer to **FIG. 20**, with the right side view being identical to the left side view, thus omitted), in top view (refer to **FIG. 21**), in bottom view (refer to **FIG. 22**). Additionally, different variant configurations of the body **10'** incorporate the approximately capsule-shaped groove portion **20'** (refer to **FIGS. 23, 24, 25**). The third embodiment may also refer to **FIGS. 26 to 32**.

[0158] Referring to **FIGS. 47 to 53**, the fourth embodiment is described as follows. Refer to **FIG. 47**: the differences from the first, second, and third embodiments lie in that the body **10''** has a uniform outer diameter extending along the axis, without division into small-diameter and large-diameter sections. This equal-diameter design provides better strength distribution. Each groove portion **20''** of the body **10''** is tapered toward both ends along the axis **10A''** and has two included angles **20A''**. The straight section **21''** remains as one, while the tapered sections **22''** corresponding to the included angles **20A''** increase to two. This results in the groove portions **20''** arranged between the two axial ends of the body **10''** forming a bi-directional tapered structure, enhancing the anti-slip effect. The single-directional tapered structure disclosed by the groove portions **20, 20'** and the bi-directional tapered structure disclosed by the groove portion **20''** can be used individually or in combination.

[0159] Considering the distribution of structural strength at both ends of the body **10, 10', 10''** of the wrench socket of the present invention, the groove portions **20, 20'** of the first, second, and third embodiments are not provided on the side surface **12A** at both ends of the body **10, 10'**. Similarly, the groove portion **20''** of the fourth embodiment of the present invention is not provided on the side surface **12A** at both ends of the body **10''**. Therefore, the groove portions **20, 20', 20''** are provided on the side surface **12A** between the two ends of the body **10, 10', 10''**, but not on the side surface at both ends. This ensures that the ends of the body **10, 10', 10''** maintain structural integrity and strength performance when subjected to torsional force.

[0160] The present invention further provides two structural designs for anti-slip and anti-rolling functions, namely an annular groove portion **30** (refer to **FIGS. 33 to 39, FIGS. 54 to 60, FIGS. 68 to 74, FIGS. 89 to 95, FIGS. 103 to 109, and FIGS. 124 to 130**) and an embossed portion **40** (refer to **FIGS. 61 to 67, FIGS. 68 to 74, FIGS. 75 to 81, FIGS. 82 to 88, FIGS. 89 to 95, FIGS. 96 to 109, FIGS. 110 to 116, and FIGS. 117 to 130**). Both can be positioned between any two groove portions **20, 20', 20''**, and can be used individually (refer to **FIGS. 33 to 39, FIGS. 54 to 60, FIGS. 61 to 67, FIGS. 75 to 81, FIGS. 82 to 88, FIGS. 96 to 102, FIGS. 110 to 116, and FIGS. 117 to 123**) or in combination (refer to **FIGS. 68 to 74, FIGS. 89 to 95, FIGS. 103 to 109, and FIGS. 124 to**

130) to achieve the optimal anti-slip and anti-rolling effects. The specific implementations of these two structures will be described in detail below.

[0161] The present invention provides at least one annular groove portion **30** positioned between any two groove portions **20**, **20'**, **20''** on the side surface **12A**. The annular groove portion **30** may be adjacent to, connected to, overlapping with, or have other equivalent positional relationships with the groove portions **20**, **20'**, **20''**. The annular groove portion **30** surrounds the side surface **12A** around the circumference of the body **10**, forming an annular structure perpendicular to the axis **10A**. The annular groove portion **30** intersects with the plurality of groove portions **20**, **20'**, **20''** on the side surface **12A**, forming an interlaced structure. When the annular groove portion **30** intersects with the groove portions **20**, **20'**, **20''**, the annular groove portion **30** may continuously pass through the groove portions **20**, **20'**, **20''**, or may be interrupted at the groove portions **20**, **20'**, **20''**, depending on the requirements of different embodiments. The number of annular groove portions **30** may be at least one, and when multiple annular groove portions **30** are provided, these annular groove portions **30** may be distributed on the side surface **12A** at equal or unequal distances. The width and depth of the annular groove portion **30** may be the same as or different from those of the groove portions **20**, **20'**, **20''**, and its cross-sectional shape may be rectangular, arc-shaped, V-shaped, U-shaped, or other geometric shapes.

[0162] The present invention provides at least one embossed portion **40** positioned between any two groove portions **20**, **20'**, **20''** on the side surface **12A**. The embossed portion **40** may be adjacent to, connected to, overlapping with, or have other equivalent positional relationships with the groove portions **20**, **20'**, **20''**. The embossed portion **40** is formed on the surface of the side surface **12A**, creating a geometric texture area composed of concave and convex structures, which significantly increases the friction for the user's hand contact. The embossed portion **40** may present cross-striped circular arrangements (refer to FIGS. **61** to **67**, FIGS. **68** to **74**, FIGS. **75** to **81**, FIGS. **82** to **88**, and FIGS. **89** to **95**), straight-striped circular arrangements (refer to FIGS. **96** to **102**, FIGS. **103** to **109**, FIGS. **110** to **116**, FIGS. **117** to **123**, and FIGS. **124** to **130**), or a matrix arrangement of multiple concave and convex geometric textures. The number of embossed portions **40** may be at least one, and when multiple embossed portions **40** are provided, these embossed portions **40** may be distributed uniformly or non-uniformly on the side surface **12A**. The depth of the embossed portion **40** may be the same as or different from that of the groove portions **20**, **20'**, **20''**, and its height, spacing, and density may vary according to the requirements of different embodiments.

[0163] In the present invention, both the annular groove portion **30** and the embossed portion **40** can be positioned between the groove portions **20**, **20'**, **20''**, and can be provided individually or in combination on the side surface **12A** according to different usage requirements and manufacturing considerations. When the annular groove portion **30** or the embossed portion **40** is used individually, each can provide anti-slip and anti-rolling effects; when the annular groove portion **30** and the embossed portion **40** are used in combination, they can form more comprehensive anti-slip and anti-rolling features. In embodiments where they are used in combination, the annular groove portion **30** and the embossed portion **40** can be arranged adjacent to each other, for example, the annular groove portion **30** can be positioned between two adjacent embossed portions **40**, or the embossed portion **40** can be positioned between two adjacent annular groove portions **30**. Alternatively, the annular groove portion **30** and the embossed portion **40** can be positioned in different areas of the side surface **12A**. The combined design of the annular groove portion **30** and the embossed portion **40** is suitable for high-precision, high-torque, or harsh environment operation requirements, significantly enhancing the convenience and safety of tool use.

[0164] In summary, the present invention provides a wrench socket which, by means of the straight section **21** and the tapered section **22** of the groove portion **20**, prevents the body **10** (refer to FIG. **5**) from rolling when placed at a workplace. When the palm with dirt (sweat or oil) grips the body **10** (refer to FIG. **6**), it prevents slipping. Thereby, operational inconvenience is avoided and

operation time is shortened.

[0165] The wrench socket of the present invention is characterized by having groove portions **20**, **20'**, **20''**, wherein the groove portions **20**, **20'**, **20''** are tapered along the axis **10A**, and form an included angle **20A** within the groove portions **20**, **20'**, **20''**, thereby providing both anti-rolling and anti-slip functions while not affecting the structural strength, offering the following technical advantages:

[0166] Efficient Force Transmission: The tapered design concentrates force along the axial direction, reducing energy loss, improving transmission efficiency, and ensuring operational stability when subjected to torsional force.

[0167] Adaptive Anti-slip: When the applied torsional force increases, fingers are guided to narrower or shallower areas, causing the anti-slip effect to gradually enhance with the intensity of applied force, improving grip stability and effectively preventing slippage.

[0168] Anti-rolling Control: The tapered design forms stable support points, effectively reducing the risk of rolling on uneven surfaces or when placed down, thereby improving operational safety.

[0169] Material Optimization: Through the design that gradually narrows in width and decreases in depth, material usage is reduced while maintaining structural strength, not only effectively reducing weight but also lowering manufacturing costs.

[0170] All documents cited in this specification are hereby individually and specifically incorporated into this specification as reference documents. To the extent that any definition or usage of a term in a cited document is inconsistent with or contradicts the definition or usage of that term in this specification, the definition in this specification shall prevail, and the definition of that term in the cited document shall not apply.

[0171] The benefits and advantages which may be provided by the present invention have been described above with regard to specific embodiments. These benefits and advantages, and any elements or limitations that may cause them to occur or to become more pronounced, are not to be construed as critical, required, or essential features of any or all of the embodiments.

[0172] The foregoing description of the preferred embodiments of the present invention has been presented for the purposes of illustration and description. It is not intended to be exhaustive or to limit the invention to the precise forms disclosed. Many modifications and variations are possible in light of the above teachings. It is intended that the scope of the invention not be limited by this detailed description, but by the claims and the equivalents to the claims appended hereto.

Claims

1. A wrench socket comprising: a body having an axis and a side surface, wherein a plurality of groove portions are arranged at intervals around the side surface, each of the plurality of groove portions is formed on the side surface between two ends of the body along the axis, and each of the plurality of groove portions is tapered along the axis and has an included angle.

2. The wrench socket as claimed in claim 1, wherein the body is provided with a small-diameter section and a large-diameter section connected to the small-diameter section along the axis, the side surface is provided on the large-diameter section of the body, the plurality of groove portions are arranged at intervals around the side surface of the large-diameter section, the body has a connecting section connecting the small-diameter section and the large-diameter section, the small-diameter section is provided with a force-receiving part from one end of the body along the axis, the large-diameter section is provided with a force-applying part from another end of the body along the axis, the force-receiving part of the small-diameter section combines with a driving member, the force-applying part of the large-diameter section combines with a driven member, wherein when force is applied to the driving member, the force-receiving part of the small-diameter section rotates under force, causing the force-applying part of the large-diameter section to apply force to the driven member, thereby causing the driven member to rotate under force.

3. The wrench socket as claimed in claim 1, wherein the side surface further defines a first end and a second end, each of the plurality of groove portions starts from the second end and terminates before the first end.
 4. The wrench socket as claimed in claim 1, wherein the side surface further defines a first end and a second end, each of the plurality of groove portions starts from the first end and terminates before the second end.
 5. The wrench socket as claimed in claim 1, wherein the side surface further defines a first end and a second end, wherein a dividing line is defined between the first end and the second end, and each of the plurality of groove portions starts from the second end and terminates at the dividing line or terminates before the dividing line.
 6. The wrench socket as claimed in claim 1, wherein the side surface further defines a first end and a second end, wherein a dividing line is defined between the first end and the second end, and each of the plurality of groove portions starts from the first end and terminates at the dividing line or terminates before the dividing line.
 7. The wrench socket as claimed in claim 1, wherein each of the plurality of groove portions is formed along the axis from a front end toward a rear end with a straight section and a tapered section connecting to the straight section, wherein the tapered section has the included angle.
 8. The wrench socket as claimed in claim 1, wherein at least one annular groove portion is provided between any two of the plurality of groove portions on the side surface.
 9. The wrench socket as claimed in claim 1, wherein at least one embossed portion is provided between any two of the plurality of groove portions on the side surface.
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